DATA ENGINEERING

Data Governance Visual Guide

Who owns the data, who can see it, and how PII stays protected.

ANIMATED WALKTHROUGH

Step through pages to play the animation

STEP 1/6

Why Governance Exists

Data without governance is a liability with a query engine.

Breach
PII exposed

Fine
GDPR/CCPA

Distrust
nobody believes reports

Governance isn't bureaucracy — it's the system that lets you say YES to data access safely, instead of defaulting to NO.

STEP 2/6

Ownership: Every Dataset Has an Owner

If everyone owns it, no one does.

Data Owner
accountable executive/team

Data Steward
day-to-day quality & definitions

Owner: decides who gets access, signs off on classification.

Steward: maintains definitions, monitors quality, triages issues.

Custodian (platform team): operates the storage and tooling.

The RACI for data: owner accountable, steward responsible, consumers informed. Write it down per domain — tribal knowledge doesn't scale.

STEP 3/6

Classification Tiers

Not all data is equal — label it, then enforce per tier.

Public

Marketing copy, published stats — no restrictions

Internal

Dashboards, aggregates — employees only

Confidential

Salaries, contracts — named access

Restricted / PII

SSN, health, biometrics — masked by default

Classification drives everything downstream: masking rules, retention, who approves access. Auto-classify with scanners; humans confirm.

STEP 4/6

Access Policies

From coarse roles to fine-grained, policy-as-code control.

RBAC

analysts → read marts; engineers → read/write staging

ABAC / row-column security

region = user's region; salary column masked

```
GRANT SELECT ON marts.orders TO ROLE analyst;

MASKING POLICY ssn_mask AS (val) RETURNS ...
```

Prefer group/role-based grants over per-user grants, and define masking in the warehouse — not in each BI tool — so policy is inescapable.

STEP 5/6

PII Handling Toolkit

Four techniques, from reversible to destructive.

Masking

Show ***-**-1234 to unauthorized roles — data stays, visibility shrinks

Tokenization

Replace with random tokens; mapping vaulted separately

Hashing

One-way for joins/analytics; salt it or it's reversible

Deletion

Right-to-be-forgotten: hard delete + purge backups on schedule

Mask by default, reveal by exception with approval and audit trail. Every unmasking should answer: who, why, and for how long.

STEP 6/6

Audit & Lineage Close the Loop

Prove who touched what, when — and where it flowed.

Access reviews recertify grants quarterly — stale access is the #1 finding in every audit. Lineage proves PII handling end-to-end for regulators.

- Log every query against restricted tables (who, when, how many rows)
- Alert on anomalous access: bulk exports, off-hours reads
- Retention policies: delete data you no longer need — you can't leak what you don't have

STEP ✓/✓

Key Takeaways

Data Governance — Remember

- Every dataset needs a named owner (accountable) and steward (day-to-day) — unowned data rots.
- Classify into tiers (public → restricted); classification drives masking, retention, and approvals.
- Grant via roles, mask in the warehouse, and prefer policy-as-code over ticket-driven access.
- PII toolkit: masking, tokenization, hashing, deletion — mask by default, reveal by exception.
- Audit logs plus lineage prove compliance; recertify access quarterly and delete what you don't need.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev